ASSESSMENT BRIEF

Programme	MSc Artificial Intelligence		
Module Title	**Deep Learning and Generative AI**		
Module Code	CMP030L043		
Module Level	7	**Assessment Type(s)**	
Word Length / Duration	**P1** 2,000 words **P2** 3,000 words	**% contribution to module mark**	**P1** 40% **P2** 60%
Deadline (date & time) for Submission	**P1** 6/3/2026,16:00 **P2** 17/4/2026,16:00	**Format/Location of submission**	
Assessment Feedback date:			

Learning Outcomes

The primary aims of the module are to:

1. Develop Expertise in Deep Learning

Equip students with an advanced understanding of deep learning principles, including neural network architectures, optimisation techniques, and model regularisation.

2. Enhance Technical Proficiency

Provide hands-on experience with frameworks such as TensorFlow and PyTorch for designing, training, and optimising deep learning and generative AI models.

3. Enable Critical Evaluation of Generative AI

Train students to critically assess generative AI models' strengths, limitations, and ethical implications, including VAEs, GANs, and diffusion models.

4. Foster Innovation in AI Applications

Encourage students to design and implement cutting-edge AI solutions that leverage deep learning and generative models for real-world challenges. 5. Promote Ethical and Scalable AI Development Ensure students understand the computational challenges, ethical considerations, and responsible AI principles involved in deploying deep learning systems at scale.	

Assessment Requirements

This module is assessed via two summative components. To successfully pass the module, **students must achieve a pass mark (50%) in BOTH Component 1 and Component 2.**

Component	Assessment Type	Weighting	Word Count	Submission Format	Deadline
Part 1	Critical Appraisal & Proposal	40%	2,000 words	PDF Document (via Turnitin)	Friday,6/3/2026 16:00PM
Part 2	Project (Artefact + Report)	60%	3,000 words	**Report:** PDF Document. **Artefact:** Link to GitHub Repository/Colab and zipped code folder (Jupyter Notebooks/Python scripts).	Friday,17/4/2026 16:00PM

Assessment Details

Part 1: Critical Appraisal & Proposal

You must select **one** research paper from the provided list or a comparable high-impact paper published within the last 3 years (subject to tutor approval). Your task is to critically analyse the paper's methodology and findings and then propose a novel improvement or extension.

Required Structure

1. **Summary (approx. 200 words):** Briefly summarize the paper's core contribution, architecture, and key results.
2. **Critical Appraisal (approx. 800 words):**
    - Evaluate the theoretical foundations (referencing module content like backpropagation, optimization, or specific architectures).
    - Critique the methodology: Are the baselines fair? Is the dataset appropriate? Are the ablation studies sufficient?

- Analyse the limitations: What does the model fail to do? What are the computational costs or ethical risks?

3. **Proposal for Improvement (approx. 1000 words):**
    - Propose a specific, technical extension to the paper (e.g., changing the loss function, introducing a new attention mechanism, applying the model to a different domain, or optimizing inference speed).
    - Justify *why* this improvement is needed based on your critique.
    - Hypothesize the expected outcome.
    - **Note:** This proposal will form the basis of your implementation in Part 2.

3. Part 2: Project (Artefact + Report)

Building on the proposal designed in Part 1, you will implement a Deep Learning or Generative AI solution. You are expected to develop a software artefact using Python (PyTorch or TensorFlow) and write a report evaluating its performance.

Requirements

1. **The Artefact (Code):**
    - Must be functional, well-commented, and reproducible.
    - Must demonstrate the implementation of the improvement/extension proposed in Part 1.
    - Must include a README.md with instructions on how to run the model.
2. **The Report (3000 words):**
    - **Methodology:** Detail how you implemented the solution. Explain the architecture, data preprocessing, and training strategy (hyperparameters, optimization).
    - **Results & Evaluation:** Present quantitative metrics (Accuracy, F1, FID, Perplexity, etc.) and qualitative analysis (visualizations, sample outputs). Compare your results against a baseline (e.g., the original paper's implementation).
    - **Critical Discussion:** Discuss challenges faced, computational constraints, and why the hypothesis from Part 1 was (or was not) supported.
    - **Ethics & Scalability:** Address the ethical implications (bias, misuse) and scalability of deploying your model in a real-world setting (referencing LO 5).

Report Format and Guidelines: The report should be approximately 8–10 pages long (A4 size, using a standard 11 or 12 pt font, single or 1.15 line spacing). This length excludes the cover

page and references (appendices are not expected, but if you include any, keep them brief). It should be well-structured and written in a clear, formal style. A recommended structure is:

- **Title:** Project title, your name/student number, module name, and submission date.
- **Abstract:** Optional. A short paragraph summarising the problem, approach, and key results.
- **Introduction:** Introduce the problem you are solving and why it is important. Clearly state the aims/objectives of your project and the scope. Provide any necessary background to set the context.
- **Background / Literature Review:** Briefly review relevant work or methods from textbooks, research papers, or industry to show your understanding of the domain. Identify if others have tackled similar problems and what approaches they used. This section demonstrates awareness of the field.
- **Methodology:** Describe your approach in detail. This includes the dataset you used (and how it was collected or sourced), data preprocessing steps, and the design of your deep learning model. Explain the architecture of your neural network, the reasoning behind your design choices, and any algorithms or techniques employed (e.g., transfer learning, regularisation, etc.). If you tried multiple approaches, describe them.
- **Experiments and Results:** Explain how you trained the model and present the results. Include details like training/validation split, training duration or epochs, and any tuning of hyperparameters. Present results with appropriate metrics (accuracy, F1-score, MSE, etc. as applicable) and use tables or graphs (learning curves, confusion matrix, sample outputs) to illustrate performance. Compare results for different model versions or against a baseline if available.
- **Discussion:** Analyse your results. Discuss what the model's performance indicates about the problem. Mention any challenges or unexpected outcomes and how you addressed them. For example, if the model had poor accuracy on certain classes or the training was unstable, provide insight into why. Relate your findings back to the objectives – did you meet them? This section is also a place to acknowledge limitations of your approach and suggest possible improvements or future work.
- **Conclusion:** Summarise what was accomplished in the project and the key takeaways. Emphasise the significance of your results and any concluding thoughts on using deep learning for this application.
- **References:** List all sources cited in the report (academic papers, online articles, datasets, libraries, etc.) in a consistent citation format (IEEE). Ensure every external idea or resource is properly referenced.

Throughout the report, **use clear headings and subheadings** to guide the reader. Numbered sections (1. Introduction, 2. Methodology, ...) are recommended. Include figure captions and table titles for any visuals, and refer to them in the text (e.g., "As shown in Figure 1…"). Writing clarity and organisation are crucial – avoid overly long paragraphs and use formal academic language.

Code and Repository Guidelines: Your code should be neatly organised and documented. If using a GitHub repo, include a README.md with instructions on how to run your code or reproduce your results. If using Colab, ensure that the notebook is well-commented and that all results (figures, printed metrics) are visible without requiring additional runs (preferably, have it run top-to-bottom). The markers will **not** significantly debug your code – it should run as provided. Using notebooks is fine, but ensure any required data files are accessible (you may include small data files in the repo or provide a script to download them). Remember to keep the repository private until submission if it's not meant to be public (you can invite the instructor/markers as collaborators or provide a private link) to avoid academic integrity issues. After grading, you can open-source your project if desired.

List of Suggested Papers:

Select **one** paper from the list below for your critique (Part 1) and implementation (Part 2).

Theme 1: The Reasoning Frontier (System 2 AI)

1. **DeepSeek-R1: Incentivizing Reasoning Capability via Reinforcement Learning**
    - *Brief:* Proposes using pure Reinforcement Learning to incentivize "System 2" reasoning (self-reflection, verification) in LLMs without human-annotated Chain-of-Thought data.
    - *Link:* arXiv:2501.12948
2. **Understanding and Benchmarking Artificial Intelligence: OpenAI's o3 Is Not AGI**
    - *Brief:* A critical analysis of OpenAI's o3 model, arguing it achieves high benchmarks through brute-force "test-time compute" search rather than genuine generalization.
    - *Link:* arXiv:2501.07458
3. **Evaluating Chain-of-Thought Monitorability**
    - *Brief:* OpenAI's framework for "reading the mind" of a model. It investigates if we can detect deception or bias by monitoring the model's internal reasoning chain.
    - *Link:*(https://openai.com/index/evaluating-chain-of-thought-monitorability/)
4. **To Mask or to Mirror: Human-AI Alignment in Collective Reasoning**
    - *Brief:* Investigates if AI agents in group settings should "mirror" human biases (for simulation fidelity) or "mask" them (for normative fairness).
    - *Link:*(https://deepmind.google/research/publications/180362/)
5. **Artificial Hivemind: The Open-Ended Homogeneity of Language Models**

- *Brief:* **(NeurIPS 2025 Outstanding Paper)**. Quantifies how RLHF training causes models to converge on a narrow, homogenized set of outputs, reducing global diversity.
- *Link:*(https://neurips.cc/virtual/2025/papers.html)

Theme 2: Architectural Evolution

6. **TRecViT: A Recurrent Video Transformer**
   - *Brief:* Reintroduces recurrent units (LRUs) for video processing, achieving 300 fps inference and matching ViT performance with 12x less memory.
   - *Link:*(https://deepmind.google/research/publications/122591/)
7. **Reversing Large Language Models for Efficient Training and Fine-Tuning**
   - *Brief:* Proposes reversible architectures for LLMs that do not need to cache activations, allowing for constant memory usage regardless of depth.
   - *Link:* arXiv:2512.02056
8. **HyperET: Efficient Training in Hyperbolic Space for Multi-modal LLMs**
   - *Brief:* Trains multimodal models in hyperbolic space to better capture hierarchical relationships (e.g., class taxonomies) with fewer parameters.
   - *Link:*(https://neurips.cc/virtual/2025/oral/118373)
9. **Memory Mosaics at Scale**
   - *Brief:* Challenges Transformers by proposing networks of associative memories that excel at storing new knowledge without catastrophic forgetting.
   - *Link:*(https://neurips.cc/virtual/2025/oral/118784)
10. **Superposition Yields Robust Neural Scaling**
    - *Brief:* A theoretical paper explaining neural scaling laws through the lens of "representation superposition" (storing more features than dimensions).
    - *Link:*(https://neurips.cc/virtual/2025/oral/116347)

Theme 3: Generative Simulation & Physics

11. **Sora 2 System Card**
    - *Brief:* Details the safety and simulation capabilities of OpenAI's Sora 2, focusing on physics simulation, audio synchronization, and post-hoc safety blocking.
    - *Link:*(https://cdn.openai.com/pdf/50d5973c-c4ff-4c2d-986f-c72b5d0ff069/sora_2_system_card.pdf)

12. **Efficiency vs. Fidelity: A Comparative Analysis of Diffusion vs. Flow Matching**
    - *Brief:* Mathematically proves why Flow Matching (Rectified Flow) is superior to Diffusion for low-resource settings due to straighter transport paths.
    - *Link:* arXiv:2511.19379
13. **InfinityStar: Unified Spacetime AutoRegressive Modeling for Visual Generation**
    - *Brief:* Proposes treating video generation as a pure autoregressive (next-token) task, challenging the dominance of diffusion models for video.
    - *Link:*(https://neurips.cc/virtual/2025/oral/117564)
14. **Representation Entanglement for Generation (REG)**
    - *Brief:* Accelerates Diffusion Transformer training by 63x by "entangling" high-level class tokens with low-level image latents.
    - *Link:*(https://neurips.cc/virtual/2025/oral/116345)
15. **Adjoint Schrödinger Bridge Sampler**
    - *Brief:* A theoretical advance in generative sampling using Schrödinger Bridges to learn transitions between arbitrary distributions without matched pairs.
    - *Link:*(https://neurips.cc/virtual/2025/oral/115787)

Theme 4: Embodied & Multimodal Intelligence

16. **ReconVLA: Reconstructive Vision-Language-Action Model**
    - *Brief:* **(AAAI 2026 Outstanding Paper)**. Improves robotic manipulation by forcing the model to reconstruct the visual scene (auxiliary task) to ensure geometric understanding.
    - *Link:* AAAI Awards
17. **Dynam3D: Dynamic Layered 3D Tokens Empower VLM for Navigation**
    - *Brief:* Uses a "patch-instance-zone" hierarchy to give navigation agents an explicit 3D memory, solving the "memoryless" problem in robotics.
    - *Link:*(https://neurips.cc/virtual/2025/oral/115716)
18. **LLM2CLIP: Powerful Language Model Unlocks Richer Cross-Modality Representation**
    - *Brief:* **(AAAI 2026 Outstanding Paper)**. Fine-tunes an LLM to act as a "teacher" for CLIP, enabling the vision model to understand complex, long captions.
    - *Link:* GitHub / Paper
19. **QoQ-Med: Building Multimodal Clinical Foundation Models**

- *Brief:* Introduces "Domain-aware Relative Policy Optimization" to solve data imbalance in medical AI, prioritizing rare diseases during training.
- *Link:*(https://neurips.cc/virtual/2025/oral/117310)

20. **CoralVQA: A Large-Scale Visual Question Answering Dataset**
    - *Brief:* A massive dataset for coral reef analysis that serves as a stress test for VLM fine-grained scientific classification capabilities.
    - *Link:*(https://neurips.cc/virtual/2025/oral/121509)

Theme 5: Theoretical Foundations

21. **Optimal Mistake Bounds for Transductive Online Learning**
    - *Brief:* **(NeurIPS 2025 Best Paper Runner-up)**. Resolves a 30-year-old open problem, proving exponential improvements in learning bounds when using transductive (unlabeled test data) settings.
    - *Link:*(https://neurips.cc/virtual/2025/oral/119099)
22. **Causal Structure Learning for Dynamical Systems (CaDyT)**
    - *Brief:* **(AAAI 2026 Outstanding Paper)**. Uses Gaussian Processes to recover true causal graphs from continuous-time data, overcoming discretization errors.
    - *Link:* arXiv:2512.14361
23. **High-Pass Matters: Sheaflet-Based Design for Hypergraph Neural Networks**
    - *Brief:* **(AAAI 2026 Outstanding Paper)**. Applies Cellular Sheaf Theory to show that high-pass filtering (detecting differences) is critical for hypergraph learning.
    - *Link:* AAAI Awards
24. **Understanding and Mitigating Numerical Sources of Nondeterminism in LLM Inference**
    - *Brief:* Reveals how floating-point non-associativity causes LLM outputs to change based on batch size/GPU count, and proposes "LayerCast" to fix it.
    - *Link:*(https://neurips.cc/virtual/2025/oral/118170)
25. **ImageNet-trained CNNs are not biased towards texture**
    - *Brief:* Challenges the famous "Texture Bias" hypothesis, showing that modern CNNs (like ConvNeXt) actually prioritize shape, similar to humans.
    - *Link:*(https://neurips.cc/virtual/2025/oral/116595)

Theme 6: Safety & Societal Alignment

26. **Gemini 3 Pro Frontier Safety Framework Report**
    - *Brief:* Details Google's "Critical Capability Levels" for assessing CBRN and Cyber risks, evaluating if the model provides "meaningful uplift" to threat actors.
    - *Link:*(https://storage.googleapis.com/deepmind-media/gemini/gemini_3_pro_fsf_report.pdf)
27. **Disempowerment Patterns in Real-World AI Usage**
    - *Brief:* A large-scale study quantifying how AI can distort user reality or values, identifying "Authority Projection" as a major risk factor.
    - *Link:*(https://www.anthropic.com/research/disempowerment-patterns)
28. **Real-time Generative AI and the Risk of Addiction**
    - *Brief:* A position paper arguing that instant, hyper-personalized AI content acts as "digital heroin," shortening dopamine loops and risking user health.
    - *Link:*(https://neurips.cc/virtual/2025/oral/126308)
29. **Generalizable Slum Detection from Satellite Imagery with Mixture-of-Experts**
    - *Brief:* **(AAAI 2026 Best Paper for Social Impact)**. Uses MoEs to map slums in low-resource cities, overcoming the domain shift problem in satellite imagery.
    - *Link:* arXiv:2511.10300
30. **PlantTraitNet: An Uncertainty-Aware Multimodal Framework**
    - *Brief:* **(AAAI 2026 Best Paper for Social Impact)**. Introduces uncertainty-aware learning to handle noisy citizen-science data for global biodiversity monitoring.
    - *Link:* AAAI Awards

Assessment Success Guidance

To excel in this assessment, follow these key tips:

- **Start Early & Plan:** Begin immediately. Break the project into manageable phases with deadlines for research, implementation, analysis, and writing.
- **Do Background Research:** Before coding, review relevant papers and articles to inform your approach and strengthen your report.

- **Justify Your Choices:** Clearly explain your design decisions in your report. Why did you choose a specific model, architecture, or hyperparameter?
- **Experiment Systematically:** Go beyond a single run. Perform hyperparameter tuning or ablation studies to demonstrate a thorough, methodical approach.
- **Visualise Your Results:** Use plots (e.g., training curves, confusion matrices) and diagrams to clearly present your findings and model architecture.
- **Analyse Results Critically:** Don't just state your results—interpret them. Discuss *why* you got them, analyse any errors, acknowledge limitations, and suggest future improvements.
- **Maintain Academic Integrity:** Your submitted work must be your own. Use AI tools responsibly for brainstorming or debugging, but never for writing your report or code. Disclose any significant AI assistance.
- **Present Professionally:** Ensure your final report is well-structured, proofread for errors, and formatted cleanly. A professional presentation makes a strong impression.
- **Use Available Resources:** Take advantage of all support, including office hours, forums, and the provided resource links.

Following this guidance can significantly improve your project's quality and your final grade.

Assessment Guidance Support, and Formative Feedback

You are not alone in this process! The teaching team is here to support you through the assessment.

Here are your avenues for help:

- **Assessment Briefing (Week 4):** We will hold a dedicated session to detail the project options, submission requirements, and answer your questions. Attendance is highly recommended.
- **Seminars & Labs:** Use lab sessions to get hands-on help from tutors, who will allocate specific time for project Q&A.
- **Moodle Q&A Forum:** Post questions on the Moodle forum, which the teaching team monitors regularly. Please do not share full solutions. Check the forum weekly for announcements and clarifications.
- **Formative Feedback:** You are strongly encouraged to submit an optional, one-page Project Update by Week 7. This is a chance to get early, ungraded feedback on your progress and direction.
- **Office Hours:** The module leader holds weekly office hours for one-on-one guidance. You can also request an appointment via the Bookings with me page.
- **Use of AI Tools:** You may use AI for support, such as brainstorming or coding assistance, but with these rules:

- **DO NOT** use AI to write your report or analysis. All final writing must be your own.
- You are responsible for testing and verifying any AI-generated code.
- Acknowledge any significant AI assistance in your submission.
- Use AI ethically to enhance your learning, not to bypass it. Ask if you are unsure.

- **Additional Support:** Please inform us early if you require special accommodations or are facing personal difficulties. The university's Academic Skills Centre is also available for writing and study support.
- **Final Feedback:** After grading, you will receive detailed feedback on your work to help you understand your performance and improve.

We encourage you to be proactive in seeking help. Engaging with these opportunities will significantly enhance your learning and performance.

Contact for Queries/ who you can contact for further information or queries

You can find information about your tutor on the module's Moodle page.

Assessment Rubrics

Rubric for Part 1: Critical Appraisal & Proposal (40%)

Criteria	**Distinction (70-100%)**	**Merit (60-69%)**	**Pass (50-59%)**	**Fail (<50%)**
Critical Analysis (40%)	Exceptional insight. Deconstructs the paper's methodology with precision. Identifies subtle flaws (e.g., data leakage, weak baselines) and links critique to broader SOTA literature.	Very good analysis. Clearly identifies strengths and weaknesses. Good understanding of the methodology. Arguments are coherent and well-justified.	Adequate analysis. Summarizes the paper well but critique is generic. Identifies obvious limitations but lacks deep technical nuance.	Descriptive only. Fails to critique; merely repeats the abstract. Misunderstands core concepts.
Proposed Improvement (30%)	Novel and feasible. The proposal demonstrates technical creativity (e.g., new loss term, architectural block) and is mathematically/theoretically justified.	Innovative. The proposal is a logical extension of the critique. It is technically sound and	Competent. Suggests a standard extension (e.g., "add more layers") without deep	Weak/Absent. Proposal is unrealistic, scientifically invalid, or missing. No

		feasible to implement.	justification. Feasibility is plausible.	logical link to the critique.
Theoretical Knowledge (20%)	Mastery of concepts. Demonstrates sophisticated understanding of deep learning theory (gradients, manifolds, attention). Terminology is precise.	Strong understanding. Concepts are applied correctly. Minor inaccuracies do not detract from the argument.	Acceptable. Basic concepts are correct, but there may be confusion regarding complex topics.	Fundamental errors. Significant gaps in understanding deep learning principles.
Academic Standards (10%)	Flawless presentation. Professional narrative. Perfect Harvard referencing. Strict adherence to word count.	Clear presentation. Well-structured. Referencing is largely correct.	Readable but disjointed. Citations exist but may have errors.	Unprofessional. Poor grammar. Missing citations or plagiarism.

Rubric for Part 2: Project (Artefact + Report) (60%)

Criteria	**Distinction (70-100%)**	**Merit (60-69%)**	**Pass (50-59%)**	**Fail (<50%)**
Technical Artefact (40%)	Professional quality. Code is optimized, modular, and reproducible. Implements complex improvements successfully. README is comprehensive.	High quality. Code runs correctly and achieves the goal. Good use of libraries (PyTorch/TF). Minor style issues.	Functional. Code works but may be inefficient or messy. Relies heavily on tutorial code with minimal modification.	Non-functional. Code crashes or is missing. Plagiarism of existing repositories.
Evaluation & Results (30%)	Rigorous. Uses advanced metrics (FID, Perplexity). Includes ablation studies and error bars. Critical analysis of *why* the model performed as it did.	Solid. Uses correct metrics. Comparison with at least one baseline. Interpretation of results is generally correct.	Basic. Reports simple metrics (accuracy) but lacks depth. No rigorous comparison.	Insufficient. No meaningful evaluation. Claims are unsubstantiated by data.
Critical Reflection (20%)	Insightful. Deep discussion of challenges (e.g.,	Good reflection. Discusses what worked/didn't.	Descriptive. Lists steps taken without	Little/No reflection. Report is a diary of tasks

	instability) and solutions. Evaluates the gap between theory and practice.	Identifies limitations of the approach.	analyzing *why*. Mentions challenges superficially.	rather than a critical analysis.
Ethics & Scalability (10%)	Sophisticated handling. Explicitly addresses specific ethical risks and computational costs. Proposes mitigations.	Competent handling. Mentions ethics/efficiency relevant to the project. Shows awareness of responsible AI.	Basic awareness. Generic mention of ethics but lacks depth. Ignores scalability.	Ignored. No discussion of ethics, safety, or scalability.

Use of Artificial Intelligence (AI)

In this section include details of the use of AI tools to support the assessment.

Please read and apply:

https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4a%20AI%20principles%20and%20guidance%20-for%20Senate.docx

https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4b%20Student%20Guidelines%20on%20the%20use%20of%20AI_.docx

The assessment is designed so that the use of AI during the assessment is possible. You must acknowledge any use of AI and appropriately cite all AI generated outputs. Please make sure you read and understand the assessment guidelines and ask your Module Leader if you have any questions.

You can find the student guidance on the use of AI in the Library and Nest

Mitigating circumstances/late penalties

Sometimes circumstances outside of your control may affect your studies and might prevent you from submitting work on time or attending an exam.

The University offers the ability for students to request additional time to complete an assessment or to defer an examination to a later date. If you are finding yourself in such a situation, please speak to your Academic Guidance Tutor, the Roehampton Student Union (RSU) or someone in the **Wellbeing team** first, who can support you. Further details can be found on the mitigating circumstances portal.

If you do not apply for or are not approved for Mitigating Circumstances, late penalties will apply. If work is submitted up to 14 days late, the mark will be capped at 40%/50% (delete as appropriate); if it is over 14 days late, it will not be marked.

Resubmissions and Reassessment

If you are required to resubmit this assessment or take part in reassessment, you will be notified via Moodle and your student email. Please ensure you check both regularly. Any reassessment tasks will follow the same learning outcomes and criteria.
Submission Checklist Before you submit, ask yourself: Have I fully answered the assessment brief? Have I met the word count and formatting requirements? Is my referencing complete and accurate? Have I declared any AI use honestly? Have I proofread my work? Am I submitting through the correct platform before the deadline?